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Course: Calculus Beta

Assignment: MyLab Math - Uge 6

Find the Taylor series generated by f at $x = a$.

$$f(x) = x^3 - 2x + 1, a = 3$$

The Taylor series generated by f at a is shown below.

$$f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \dots$$

First find $f(a)$ if $a = 3$.

$$f(x) = x^3 - 2x + 1$$

$$\begin{aligned} f(3) &= (3)^3 - 2(3) + 1 \\ &= 22 \end{aligned}$$

Now find $f'(x)$. Differentiate $f(x)$ with respect to x .

$$f(x) = x^3 - 2x + 1$$

$$f'(x) = 3x^2 - 2$$

Evaluate $f'(3)$.

$$f'(x) = 3x^2 - 2$$

$$\begin{aligned} f'(3) &= 3(3)^2 - 2 \\ &= 25 \end{aligned}$$

Now find $f''(x)$. Differentiate $f'(x)$ with respect to x .

$$f'(x) = 3x^2 - 2$$

$$f''(x) = 6x$$

Evaluate $f''(3)$.

$$f''(x) = 6x$$

$$\begin{aligned} f''(3) &= 6(3) \\ &= 18 \end{aligned}$$

Find $f'''(x)$. Differentiate $f''(x)$ with respect to x .

$$f''(x) = 6x$$

$$f'''(x) = 6$$

Evaluate $f'''(3)$ if $f'''(x) = 6$. Therefore, $f'''(3) = 6$.

Notice that $x^3 - 2x + 1$ has an order of 3; therefore, $x^3 - 2x + 1$ has non-zero derivatives up to the third derivative. The fourth derivative and beyond are equal to zero and thus can be eliminated from the Taylor series which becomes

$$f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + 0 + 0 + \dots$$

To find the Taylor series generated by $x^3 - 2x + 1$ at $a = 3$, substitute 22 for $f(3)$, 25 for $f'(3)$, 18 for $f''(3)$, and 6 for $f'''(3)$ into the equation below.

$$f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3$$

$$22 + 25(x - 3) + \frac{18}{2!}(x - 3)^2 + \frac{6}{3!}(x - 3)^3$$

Simplifying the expression in the previous step gives the Taylor series generated by f .

$$22 + 25(x - 3) + 9(x - 3)^2 + (x - 3)^3$$